IP Packet Format

IP Packets Product Documentation contains the main data transmission protocol used in a variety of packet formats, packet example. By IP Packets Product Documentation, be able to enhance understanding of various protocol packets.

Data Link Layer
MPLS Layer
Network Layer
Transport Layer
TCP
UDP Packet Format
SCTP

Search this document. 0 / 500

Please evaluate and leave your comments:

TCP

The Transmission Control Protocol (TCP) is intended for use as a highly reliable host-to-host protocol between hosts in packet-switched computer communication networks, and in interconnected systems of such networks.

Packet Format

Figure 4-1 TCP Header Format

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|          Source Port          |       Destination Port        |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                        Sequence Number                        |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                    Acknowledgment Number                      |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|  Data |           |U|A|P|R|S|F|                               |
| Offset| Reserved  |R|C|S|S|Y|I|            Window             |
|       |           |G|K|H|T|N|N|                               |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|           Checksum            |         Urgent Pointer        |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                    Options                    |    Padding    |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
|                             data                              |
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+
```

Field	Length	Description
Source Port	16 bits	The source port number.
Destination Port	16 bits	The destination port number.
Sequence Number	32 bits	The sequence number of the first data octet in this segment (except when SYN is present). If SYN is present the sequence number is the initial sequence number (ISN) and the first data octet is ISN+1.
Acknowledgment Number	32 bits	If the ACK control bit is set this field contains the value of the next

Translation
Favorite
Download
Update Date : 2026-02-02
Related Documents
Digital Signature File
Digital Signature Tool
Share

Previous Next

		this is always sent.
Data Offset	4 bits	The number of 32 bit words in the TCP Header. This indicates where the data begins. The TCP header (even one including options) is an integral number of 32 bits long.
Reserved	6 bits	Reserved for future use. Must be zero.
URG	1 bit	Urgent Pointer field significant
ACK	1 bit	Acknowledgment field significant
PSH	1 bit	Push Function
RST	1 bit	Reset the connection
SYN	1 bit	Synchronize sequence numbers
FIN	1 bit	No more data from sender
Window	16 bits	The number of data octets beginning with the one indicated in the acknowledgment field which the sender of this segment is willing to accept.
Checksum	16 bits	The checksum field is the 16 bit one's complement of the one's complement sum of all 16 bit words in the header and text. If a segment contains an odd number of header and text octets to be checksummed, the last octet is padded on the right with zeros to form a 16 bit word for checksum purposes. The pad is not transmitted as part of the segment. While computing the checksum, the checksum field itself is replaced with zeros.
Urgent Pointer	16 bits	This field communicates the current value of the urgent pointer as a positive offset from the sequence number in this segment. The urgent pointer points to the sequence number of the octet following the urgent data. This field is only be interpreted in segments with the URG control bit set.
Options	variable	Options may occupy space at the end of the TCP header and are a multiple of 8 bits in length. All options are included in the checksum. An option may begin on any octet boundary. There are two cases for the format of an option: • Case 1: A single octet of option-kind. • Case 2: An octet of option-kind, an octet of option-length, and the

Previous Next

		octets of option-kind and option-length as well as the option-data octets. Note that the list of options may be shorter than the data offset field might imply. The content of the header beyond the End-of-Option option must be header padding (i.e., zero).
Padding	variable	The TCP header padding is used to ensure that the TCP header ends and data begins on a 32 bit boundary. The padding is composed of zeros.
data	variable	Payload.

Packet Example

Figure 4-2 TCP Segment

```
Frame 1: 153 bytes on wire (1224 bits), 153 bytes captured (1224 bits)
    Encapsulation type: Ethernet (1)
    Arrival Time: May 12, 2015 17:29:30.289263000
    [Time shift for this packet: 0.000000000 seconds]
    Epoch Time: 1431422970.289263000 seconds
    [Time delta from previous captured frame: 0.000000000 seconds]
    [Time delta from previous displayed frame: 0.000000000 seconds]
    [Time since reference or first frame: 0.000000000 seconds]
    Frame Number: 1
    Frame Length: 153 bytes (1224 bits)
    Capture Length: 153 bytes (1224 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:telnet]
    [Coloring Rule Name: TCP]
    [Coloring Rule String: tcp]
Ethernet II, Src: 18:c5:8a:17:b6:b3 (18:c5:8a:17:b6:b3), Dst: 90:e2:ba:96:08:07 (90:e
2:ba:96:08:07)
    Destination: 90:e2:ba:96:08:07 (90:e2:ba:96:08:07)
        Address: 90:e2:ba:96:08:07 (90:e2:ba:96:08:07)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Source: 18:c5:8a:17:b6:b3 (18:c5:8a:17:b6:b3)
        Address: 18:c5:8a:17:b6:b3 (18:c5:8a:17:b6:b3)
        .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
        .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
    Type: IP (0x0800)
Internet Protocol Version 4, Src: 10.10.144.1 (10.10.144.1), Dst: 192.168.128.110 (19
2.168.128.110)
    Version: 4
    Header Length: 20 bytes
    Differentiated Services Field: 0xc0 (DSCP 0x30: Class Selector 6; ECN: 0x00: No
t-ECT (Not ECN-Capable Transport))
        1100 00.. = Differentiated Services Codepoint: Class Selector 6 (0x30)
        .... ..00 = Explicit Congestion Notification: Not-ECT (Not ECN-Capable Transpo
rt) (0x00)
    Total Length: 139
    Identification: 0x19f5 (6645)
    Flags: 0x00
        0... .... = Reserved bit: Not set
        .0.. .... = Don't fragment: Not set
        ..0. .... = More fragments: Not set
    Fragment offset: 0
    Time to
```

Previous Next

```
    [Good: False]
    [Bad: False]
  Source: 10.10.144.1 (10.10.144.1)
  Destination: 192.168.128.110 (192.168.128.110)
  [Source GeoIP: Unknown]
  [Destination GeoIP: Unknown]
Transmission Control Protocol, Src Port: telnet (23), Dst Port: 63217 (63217), Seq: 1,
Ack: 1, Len: 99
  Source Port: telnet (23)
  Destination Port: 63217 (63217)
  Sequence number: 1    (relative sequence number)
  Acknowledgment number: 1    (relative ack number)
  Header Length: 20 bytes
  .... 0000 0001 1000 = Flags: 0x018 (PSH, ACK)
    000. .... .... = Reserved: Not set
    ...0 .... .... = Nonce: Not set
     .... 0... .... = Congestion Window Reduced (CWR): Not set
    .... .0.. .... = ECN-Echo: Not set
    .... ..0. .... = Urgent: Not set
    .... .... 1... = Push: Set
    .... .... .0.. = Reset: Not set
    .... .... ..0. = Syn: Not set
    .... .... ...0 = Fin: Not set
  Window size value: 40960
  [Calculated window size: 40960]
  [Window size scaling factor: -1 (unknown)]
  Checksum: 0xed6d [validation disabled]
    [Good Checksum: False]
    [Bad Checksum: False]
  Urgent pointer: 0
  [SEQ/ACK analysis]
    [Bytes in flight: 99]
Telnet
  Data: \r\n
  Data: Info: The max number of VTY users is 10, and the number\r\n
  Data:       of current VTY users on line is 1.
```

Reference

Standard	Description
RFC 9293	Transmission Control Protocol

Protocol Layering

Figure 4-3 TCP Protocol Layering

```
+---------------------+
|    higher-level     |
+---------------------+
|        TCP          |
+---------------------+
| internet protocol   |
+---------------------+
|communication network|
+---------------------+
```

HUAWEI

Quick Links

Huawei e+ App

HUAWEI eKit App

Huawei HiKnow APP

HUAWEI eFly App

Copyright © 2026 Huawei Technologies Co., Ltd. All rights reserved.

Privacy Terms of use Cookies Cookie Settings Report content

Previous Next